

**‘Checkpoint Report – Broughty Ferry Media: Customer Renewal
Prediction’**



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1.0 Introduction

Broughty Ferry Media is a UK-based magazine publisher that is seeking to maximise its return on a £20,000 renewal campaign. With a contact cost of £40 per customer and renewal revenue of £140, a maximum of 500 customers can be targeted by the campaign ($£20,000 / £40$). The publisher's business objective is therefore to identify the 500 customers who are more likely to renew their subscriptions compared to others so that the renewal campaign generates a positive return.

This report will aim to cover the methods of the training/validation split of the data, as well as an exploratory analysis of the dataset to help select the top 10 variables that can predict the probability of renewal with the best accuracy. Once these predictor variables have been selected, three models will be constructed on the training set and compared on the validation set to determine the model that will most likely generalise well to unseen data.

2.0 Data Split (Training/Validation)

The observed dataset, consisting of 8,000 records and 38 variables, was divided into a training set (70%, $n \approx 5,600$ records) and a validation set (30%, $n \approx 2,400$ records) using stratified random sampling. This was achieved by using the function `createDataPartition()` with `set.seed(2417549)`. By employing stratified random sampling, the proportion of renewals is equal across both the training and validation sets, preventing the occurrence of class imbalance from distorting accurate model evaluation. The renewal rate in both the training set and validation set was 6.25%, confirming that stratification was successful. The validation set acts as a sample to use in comparing the models built on the training data prior to receiving the test data, allowing for an unbiased evaluation.

3.0 Exploratory Data Analysis

3.1 Renewal Rate

An initial review of the target variable (Renewal) shows that 500 customers (approximately 6.25%) renewed their subscription, while the remaining 7500 (93.75%) did not. This class imbalance means that if a model predicted that no customer would renew, it would still appear to be 93.75% accurate. This highlights that accuracy alone cannot be relied upon when evaluating model performance, and other metrics, such as AUC, may prove to be more suitable.

3.2 Missing Values

To ensure that the data can be appropriately used, a systematic check for missing values was conducted across all 38 variables. It was found that there were no missing values in the dataset, so no further strategies were required.

3.3 Key Variable Distributions

Exploratory boxplots and bar charts were produced for several candidate variables against the renewal outcome to help determine if there was any correlation between a variable and the likelihood of the customer renewing their subscription. The following candidate variables were explored: renewal rate by active subscription (Figure 1a), total amount paid by renewal (Figure 1b), years since last order (Figure 1c), and paid direct mail orders (Figure 1d). For example, Figure 1a shows that customers with more active subscriptions showed higher renewal rates, suggesting that current engagement may be a useful predictor of renewal intent. Moreover, Figure 1b illustrates that customers with a higher TotalAmountPaid value have a more strongly associated relationship with renewal, suggesting that total spend may reflect loyalty in customers.

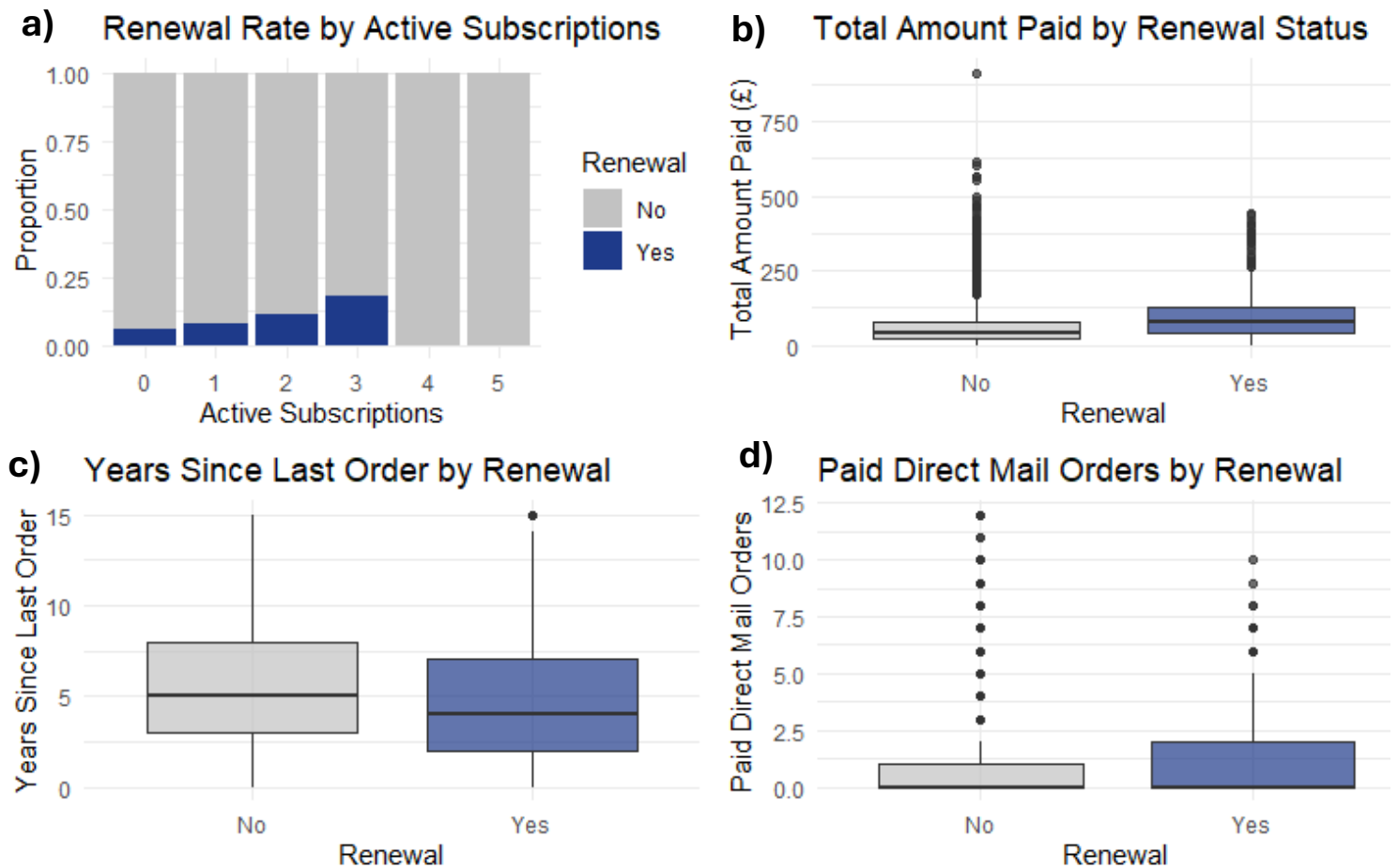


Figure 1. Visuals for Exploratory Data Analysis. (a) Renewal rate by number of active subscriptions; customers with more active subscriptions demonstrate a higher likelihood to renew. (b) Distribution of TotalAmountPaid by renewal status; customers who renewed show a higher median, suggesting monetary engagement is a reliable indicator of renewal intent. (c) Years since Last Order by Renewal status; customers who renewed show a lower median, suggesting more recent engagement. (d) Paid Direct Mail Orders by renewal status; renewers demonstrate slightly higher direct mail responsiveness, consistent with habitual purchasing behaviour.

3.4 Correlation with Renewal

In addition to exploratory visual inspection, quantitative analysis was also performed on the dataset to aid in variable selection. To initially determine what variables should be selected, point-biserial correlation calculations were performed between all the numeric predictors and the binary Renewal outcome (Table 1). The results of this analysis show that PoundsPerIssue has the strongest linear association with renewal, followed by TotalAmountPaid and Income, providing a strong foundation for variable selection. The correlation values are moderate in magnitude, which is to be expected given that no single variable can perfectly predict renewal behaviour.

Variable	Point-Biserial Correlation with Renewal
PoundsPerIssue	0.2017
TotalAmountPaid	0.1650
Income	0.1641
MonthsSince1stOrder	0.1549
Child6.12	0.1527
TotalPaidOrders	0.1440
Child13.18	0.1188
HouseholdSize	0.1015
PaidCashMagazines	0.0895
HomeValue	0.0786

Table 1. Top 10 numeric predictors ranked by point-biserial correlation with renewal.

In addition to the numeric variables examined above, several categorical variables (e.g. MagazineStatus, LastPaymentType, and GiftDonor) were reviewed, however, since point-biserial correlation requires a numeric predictor, a suitable comparison couldn’t be conducted. Given that these variables would require dummy encoding before testing and the top ten numeric predictors already covered a broad range of dimensions, they were excluded from the final variable selection.

4.0 Variable Selection

4.1 Selected Variables and Justification

From the 38 available variables, ten were selected based on their theoretical relevance to subscription renewal and statistical association with the target variable (Table 2). The point-biserial correlations that were calculated in Section 3.4 provided a quantitative ranking of each variable’s linear association with the target, with correlations ranging from $r = 0.2017$ (PoundsPerIssue) to $r = 0.0786$ (HomeValue), confirming the selection was data-driven rather than based on assumption. Additionally, each variable was evaluated on its theoretical relevance to renewal behaviour, drawing on the Recency, Frequency, and Monetary (RFM) framework. Together, these variables provide broad coverage across the key dimensions of customer engagement.

Variable	Type	Rationale
PoundsPerIssue	Numeric	Strongest predictor: spend per issue reflects engagement value
TotalAmountPaid	Numeric	Cumulative monetary spend signals long-term engagement and loyalty
Income	Numeric	Financial capacity to renew; higher-income customers are less price-sensitive
MonthsSince1stOrder	Numeric	Longer relationships often correlate with higher loyalty
Child6.12	Numeric	Household composition influences subscription habits as a key magazine-reading demographic

TotalPaidOrders	Numeric	Order frequency reflects habitual purchasing behaviour
Child13.18	Numeric	Household composition; older children may drive magazine interest
HouseholdSize	Numeric	Larger households may have multiple readers, supporting renewal
PaidCashMagazines	Numeric	Cash payment behaviour indicates active and committed engagement
HomeValue	Numeric	Factor indicating financial stability and disposable income

Table 2. Selected predictor variables and justification for inclusion.

4.2 Multicollinearity – VIF Test

Following variable selection, a Variance Inflation Factor (VIF) test was conducted to determine if any of the ten selected predictors were redundant and collinear with one another. If the predictors are shown to be highly correlated, it means their individual coefficient estimates become “unstable”, ultimately undermining the reliability of significance tests and the interpretation of odds ratio. VIF tests can quantify this collinearity with a value of 1 indicating no collinearity at all and a value of 10 suggesting severe collinearity. The VIF was calculated by fitting a logistic regression model on the ten selected variables and extracting the inflation factor for each predictor (Table 3). All ten variables returned VIF values below 5, with the highest being TotalPaidOrders (VIF = 3.173) and TotalAmountPaid (VIF = 3.003), which is to be expected as they both capture purchasing behaviour, so they will share some variance. The absence of severe multicollinearity across the ten variables makes them appropriate for use across the three created models.

Variable	VIF Value
PoundsPerIssue	1.209
Income	1.636
Child6.12	1.276
TotalAmountPaid	3.003
MonthsSince1stOrder	1.559
TotalPaidOrders	3.174
Child13.18	1.308
PaidCashMagazines	1.439
HouseholdSize	1.219
HomeValue	1.585

Table 3. Variance Inflation Factors (VIF) for all 10 selected predictor variables.

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It should be noted that VIF is most directly applicable to a logistic regression model, as multicollinearity would destabilise coefficient estimates. Tree-based methods, on the other hand, are inherently more robust to correlated predictors as they select features at each split independently. However, confirming that all selected variables have low VIF provides additional confidence that no predictor is entirely redundant, benefitting the overall quality of the variable selection.

5.0 Model Building

Regarding the selection of a model suitable for Broughty Ferry Media’s campaign, three models were created, trained on the appropriate set, and evaluated against various dimensions. To determine the models’ performance, they were assessed on accuracy, sensitivity (recall), specificity, precision, and Area Under the ROC Curve (AUC). The primary selector of these factors was the AUC performance, as this evaluates the discrimination ability of the model across all possible classification thresholds. These dimensions allow for a more informed decision when compared to relying on accuracy alone, something that is beneficial when the business needs to rank customers by renewal probability.

5.1.1 Logistic Regression

When creating models with the focus on binary classification, logistic regression is the natural baseline. This models the log-odds of the target variable as a linear combination of the selected predictors, producing probability estimations. Coefficients can be seen as directly interpretable as log-odds multipliers with significant coefficients ($p < 0.05$) providing statistical evidence of association with renewal. The two strongest significant predictors were PoundsPerIssue ($OR > 1$, $p < 0.05$) and Income ($OR > 1$, $p < 0.05$), confirming that higher spend per issue and greater financial capacity increase the odds of renewal.

5.1.2 Decision Tree

A classification tree (CART) was fitted with a complexity parameter of 0.001 and a maximum depth of 6 to balance model complexity against interpretability. Decision trees can produce explicit if-then rules that are easily understandable to business stakeholders. However, it is important to note that lower CP values permit more complex trees but increase the risk of overfitting. That is why a value of 0.001 was selected to capture non-linear interactions while avoiding a tree so complex it would memorise noise.

5.1.3 Random Forest

A Random Forest of 500 trees was created and trained with the `mtry` set to 3 (the square root of the number of predictors) to allow for appropriate randomisation. Random Forests achieve this by randomising the set of features that are considered at each split, in turn reducing the correlation between trees, and typically producing lower generalisation error than a single decision tree. The inclusion of the variable importance plots helps to identify which predictors contribute the most to renewal probability, providing a secondary source of insight to aid in business interpretation (Figure 2). Figure 2 presents the variable importance output from the Random Forest created from the dataset (`set.seed(2417549)`), measured by the Mean Decrease in Gini Impurity. PoundsPerIssue was shown to be the dominant predictor, consistent with the highest point-biserial correlation identified in Section 3. Importantly, the variable importance ranking provides additional evidence for the variable selection decisions in Section 4, with variables selected demonstrating importance in the models.

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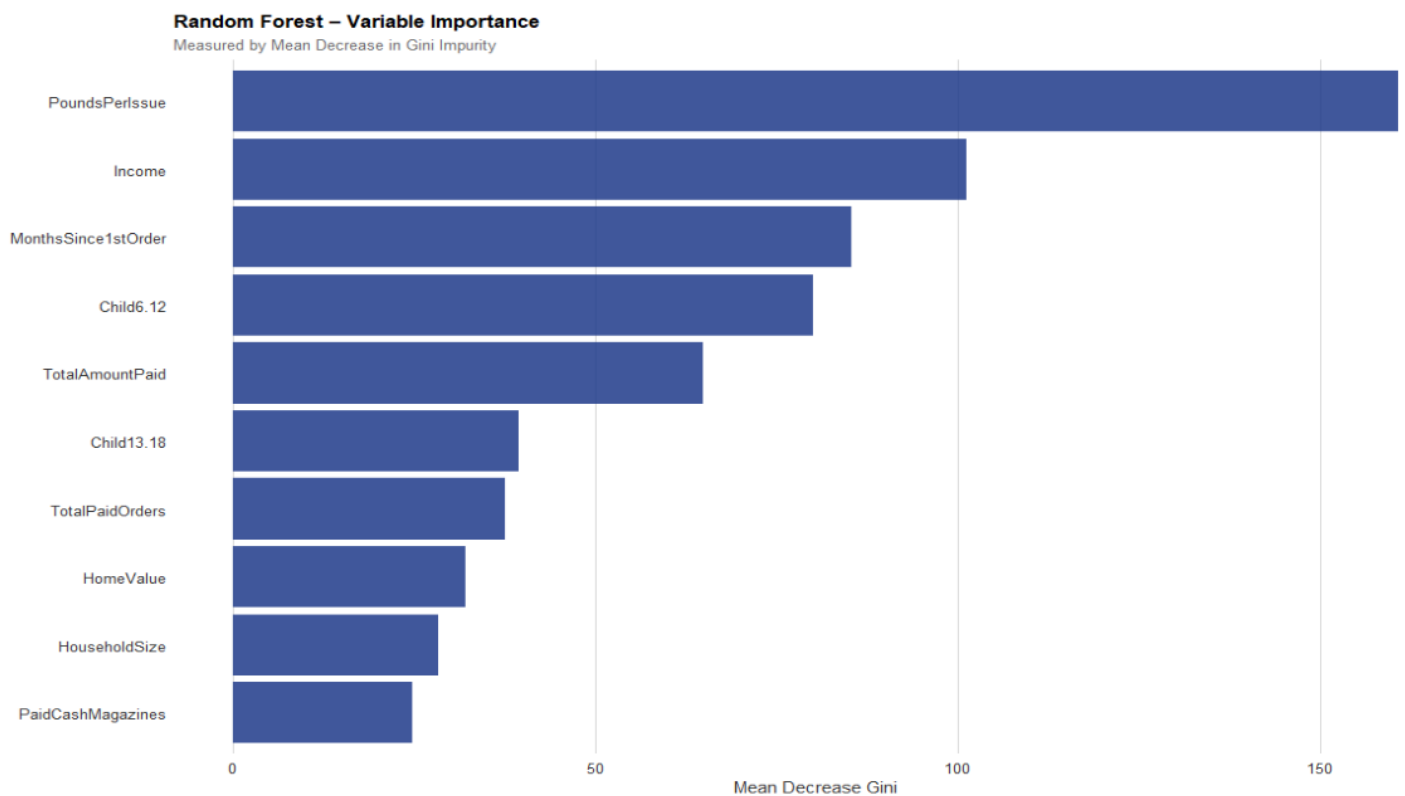


Figure 2. Random Forest variable importance ranked by Mean Decrease in Gini Impurity. Higher values indicate greater contribution to reducing classification error across all 500 trees.

6.0 Model Comparison and Selection

6.1 ROC Curves

ROC curves for all three of the models were plotted on the same axes (Figure 3) to allow for a more direct visual comparison. A curve closer to the top-left corner indicates better performance as it achieves higher sensitivity without sacrificing specificity to do so. The Random Forest curve sits consistently higher than both the Decision Tree and Logistic Regression, as seen by its higher AUC value of 0.8903, confirming that it is the strongest model in distinguishing customers likely to renew from those who are not.

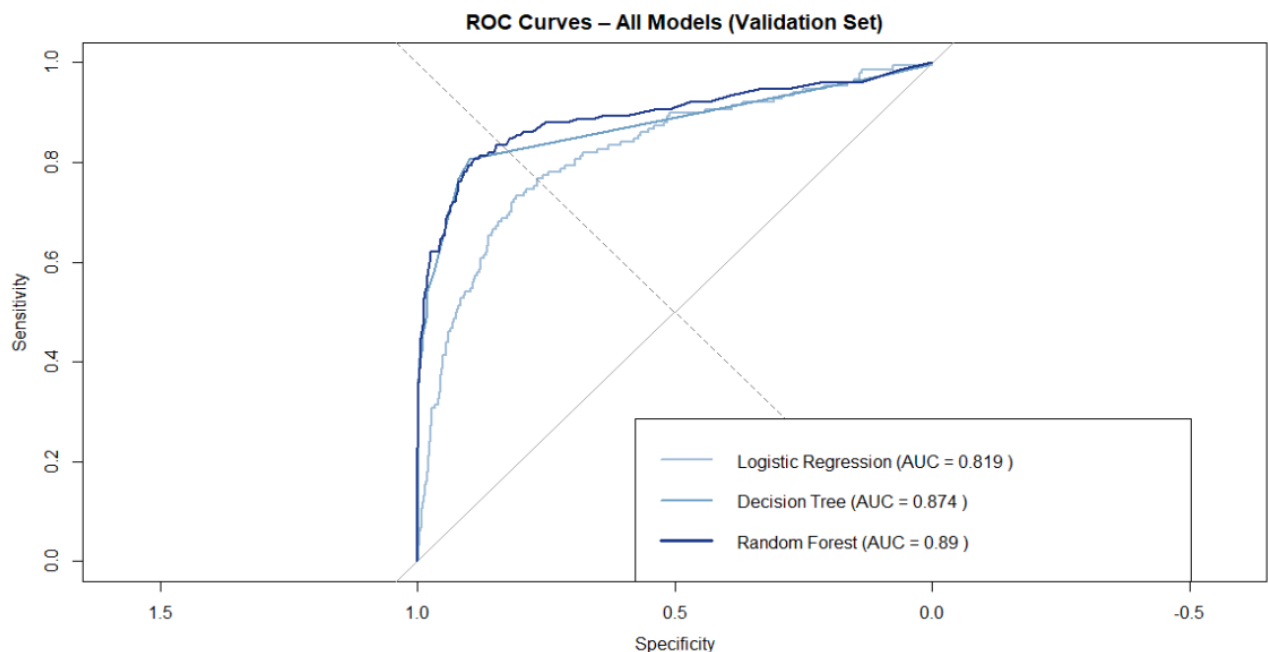


Figure 3. ROC Curves for all three models on the validation set. AUC values are shown in the legend, and the dashed diagonal line represents the random chance (AUC = 0.5).

6.2 Performance Metrics

The table below (Table 4) summarises the validation-set performance of all three models, with their AUC values indicating the probability that a randomly selected renewing customer is ranked above a randomly selected non-renewing customer. Precision (the proportion of customers predicted to renew who actually did) is also included, as it is directly relevant to how efficient the campaign is. A higher precision value means fewer wasted contacts and a better return on the £40 contact cost.

Model	Accuracy	Sensitivity	Specificity	Precision	AUC
Logistic Regression	0.9362	0.0667	0.9942	0.4348	0.8186
Decision Tree	0.9517	0.4867	0.9827	0.6518	0.8740
Random Forest	0.9571	0.3933	0.9947	0.8310	0.8903

Table 4. Predictive performance metrics for all three models on the validation set. The highlighted row indicates the selected model.

6.3 Profit Simulation and Selected Model

In addition to calculating the statistical metrics of the models, a business-aligned profit simulation was performed to rank customers based on their predicted renewal probability and to calculate the estimated profit from targeting the top-n customers. As the validation set represents 30% of the full dataset (2,400 of 8,000 customers), the campaign budget was scaled proportionally to 150 contacts (500×0.30) to ensure that the simulation reflects the correct ratio relative to the dataset. The calculation, however, still applies the business parameters of £40 per contact and £140 revenue per renewal.

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The simulation revealed that the Random Forest outperformed both the Decision Tree and the Logistic Regression models on profit generation, producing £6,740 from 91 renewals across the top 150 targeted customers (Table 5). All three of the models substantially outperform the random selection baseline which produced a loss of £4,740 from 9 renewals, confirming that the modelling approach delivers better business value over untargeted campaigning.

Based on these results, the Random Forest is selected as the final model as it provides a superior AUC (0.8903 vs. 0.8740 for the Decision Tree) and its highest simulated profit on the validation set. These two measurements, one statistical and one business-aligned, provide strong evidence that the Random Forest generalises well and is the most appropriate model for the campaign.

Model	Customers Targeted	Renewals	Cost (£)	Revenue (£)	Profit (£)
Random Selection (Baseline)	150	9	6000	1260	-4740
Logistic Regression	150	51	6000	7140	1140
Decision Tree	150	86	6000	12040	6040
Random Forest	150	91	6000	12740	6740

Table 5. Profit simulation metrics for all three models on the validation set for the top 150 customers. The highlighted row indicates the model with the highest profit.

7.0 Conclusion

This report has established a robust analytical pipeline for predicting magazine subscription renewals at Broughty Ferry Media. For future analysis, ten predictor variables have been selected based on their theoretical relevance and statistical performance, with them covering various dimensions of customer engagement, recency, frequency, and monetary value. From the training dataset, three models were trained (Logistic Regression, Decision Tree, and Random Forest) and compared using a validation set, with the Random Forest model being selected as the final model based on its AUC values and business profit simulation performance.

Once the test data has been released, the Random Forest model will be applied to generate renewal probabilities for all the test customers featured. These probabilities will then be used to rank customers, identify the optimal targeting strategy, and produce a full profit analysis and business recommendation.